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22CH203 - ENGINEERING CHEMISTRY II

UNIT III – METAL CORROSION AND ITS PREVENTION

Lesson Plan 1.2.
Topic: Dry or Chemical Corrosion

References: 1. Chemistry for Engineering Students (Lawrence S. Brown, Thomas A. Holme), 4th Edition- Chapter 13.
2. Inorganic Chemistry (Gary L.Miessler Donald A.Tarr), 3rd Edition- Chapter 8.

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1. CLASSIFICATION / THEORIES / TYPES OF CORROSION:

Based on the environment, corrosion is classified into Dry or Chemical corrosion, and Wet or Electrochemical corrosion.

1.1. Dry or Chemical Corrosion:

Dry corrosion is due to the attack of metal surfaces by atmospheric gases such as oxygen, hydrogen sulphide, sulphur dioxide, nitrogen, etc.

There are 3 main types of dry corrosion.

(i) Oxidation corrosion (or) corrosion by oxygen.
(ii) Corrosion by hydrogen
(iii) Liquid-metal corrosion

1.1.1. Oxidation Corrosion or Corrosion by Oxygen:

Oxidation corrosion is brought about by the direct attack of oxygen at low or high temperatures on metal surfaces in the absence of moisture.

Examples:

- Alkali metals (Li, Na, K, etc.) and alkaline-earth metals (Mg, Ca, Sn, etc.) are rapidly oxidized at low temperatures.
- At high temperatures, almost all metals (except, Ag, Au and Pt) are oxidized.

Mechanism:

- Oxidation initially occurs at the surface of the metal resulting in the formation of metal ions(M^{2+}) which occurs at the metal/oxide interface.

$$M \rightarrow M^{n+} + ne^-$$

- Oxygen changes to ionic form (O^{2-}) due to the transfer of electrons from metal, which occurs at the oxide film/environment interface.

$$1/2\ O_2 + 2e^- \rightarrow O^{2-}$$

- Oxide ions reacts with the metal ion to form the metal oxide film

$$M + \tfrac{1}{2}O_2 \dashrightarrow M^{2+} + O^{2-} \equiv MO \text{ (Metal-oxide film).}$$

- Once the metal surface is converted to a monolayer of metal-oxide, for further corrosion(oxidation) to occur, the metal ion diffuses outward through the metal-oxide barrier. Thus, the growth of oxide film commences perpendicular to the metal surface.

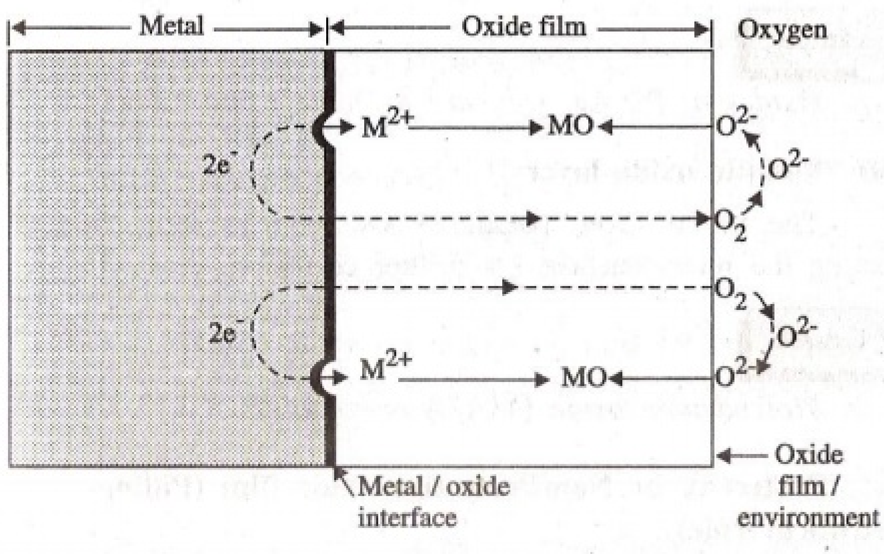

Figure 1. Mechanism of Oxidation Corrosion.

1.1.2. Nature of Oxide Film:

The nature of oxide film formed on the metal surface plays an important role in oxidation corrosion. Based on the nature of metal used, oxide layers formed are of following types

(i) Stable oxide layer:

A stable oxide layer is a fine-grained structure, and gets adsorbed tightly to the metal surface. Such a layer is impervious in nature and stops further oxygen attack through diffusion.Such a film behaves as a protective coating and no further corrosion can develop.

Examples are oxides of Al, Sn, Pb, Cu etc.

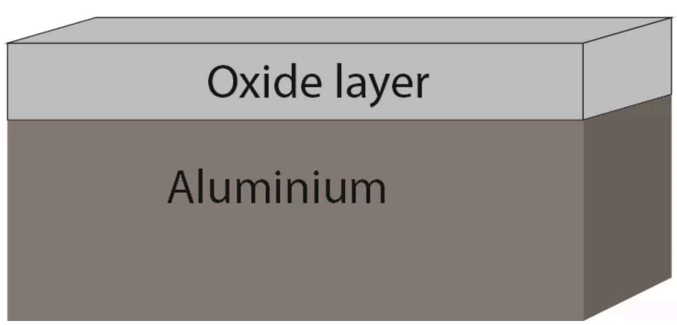

Figure 2. Aluminium oxide layer protection of Aluminium.

(ii) Unstable oxide layer:

Unstable oxide layer is mainly produced on the surface of noble metals, which decomposes back into the metal and oxygen.

Example: Oxides of Pt, Ag etc.

(iii) Volatile oxide layer:

The oxide layer volatilizes as soon as it is formed, leaving the metal surface for further corrosion. Molybdenum oxide (MoO_3) is volatile.

(iv) Porous oxide layer:

The oxide layer formed having a number of pores and cracks leading to rapid corrosion.

Example: Alkali and alkaline earth metals.

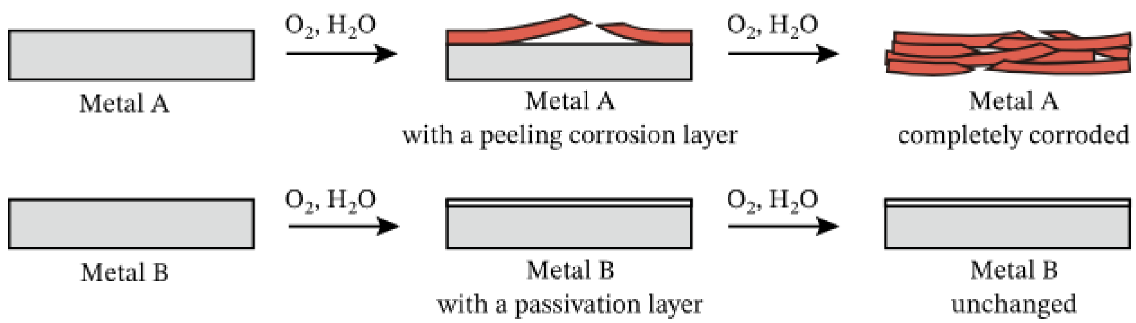

Figure 3. The role of the oxide layer in corrosion prevention. (Metal A: corroded; Metal B: passivation layer prevents further corrosion)

1.1.3. Pilling Bedworth Rule:

According to the Pilling-Bedworth rule, if the volume of the oxide layer formed is less than the volume of the metal, the oxide layer is porous and non-protective. The volume of oxides of alkali and alkaline earth metals such as Na, Mg, Ca, etc., is less than the volume of the metal consumed. Hence the oxide layer formed is porous and non-protective.

On the other hand, if the volume of the oxide layer formed is greater than the volume of the metal, the oxide layer is non-porous and protective.

$$PB = \frac{\text{volume of oxide produced}}{\text{volume of metal consumed}}$$

P-B Ratio	Oxide Layer Properties		Applications
	Porosity	Protective	
$PB \geq 1$	Non-Porous	Protective	Preventing further corrosion.
$PB < 1$	Porous	Non-Protective	Continued corrosion.

Pilling Bed worth Ratio (PB):

The ratio of the volume of the metal oxide formed to the volume of the metal consumed is called Pilling- Bed worth ratio. If the PB ratio is equal to or greater than 1, the metal oxide layer will be protective and non- porous. If the PB ratio is less than 1, the metal oxide layer will be non-protective and porous (as summarized in Table).

Oxide	Oxide/metal volume ratio
K_2O	0.45
MgO	0.81
Na_2O	0.97
Al_2O_3	1.28
ThO_2	1.30
ZrO_2	1.56
Cu_2O	1.64
NiO	1.65
FeO (on α-Fe)	1.68
TiO_2	1.70–1.78
CoO	1.86
Cr_2O_3	2.07
Fe_3O_4 (on α-Fe)	2.10
Fe_2O_3 (on α-Fe)	2.14
Ta_2O_5	2.50
Nb_2O_5	2.68
V_2O_5	3.19
WoO_3	3.30

Table 1. Oxide-Metal Volume Ratios of Some Common Metals.

1.1.4. Corrosion by Hydrogen:

Hydrogen Embrittlement (at ordinary temperature):

When metals contact H_2S at ordinary temperature it causes the evolution of atomic hydrogen.

$$Fe + H_2S \text{----}\!> FeS + 2H$$

This atomic hydrogen diffuses readily into the metal and collects in the voids, where it recombines to form molecular hydrogen.

$$H + H \text{----}\!> H_2\uparrow$$

Collection of these hydrogen gases in the voids develops very high pressure, which causes cracks and blisters on metal. Thus, the process of formation of cracks and blisters on the metal surface, due to high pressure of hydrogen gas is called hydrogen embrittlement.

Decarburisation:

At higher temperatures, atomic hydrogen is formed by the thermal dissociation of molecular hydrogen.

$$H_2 \xrightarrow{\text{Heat}} 2H$$

When steel is exposed to this environment, the atomic hydrogen readily combines with the carbon of steel and produces methane gas.

The collection of these gases in the voids develops very high pressure, which causes cracking. Thus the

$$C + 4H \longrightarrow CH_4\uparrow$$

process of decrease in carbon content in steel is termed as "decarburisation" of steel.

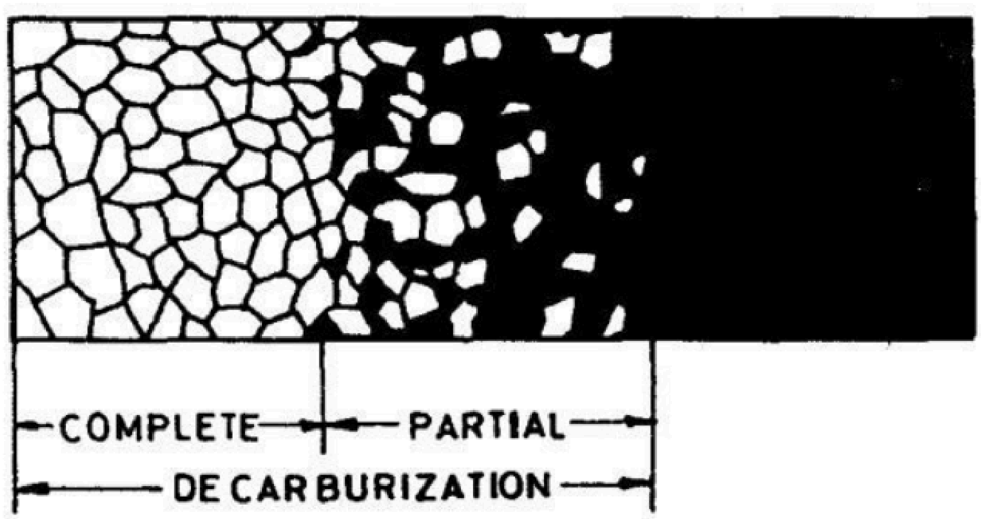

Figure 4. Partial and Complete Decarburization.

1.1.5. LIQUID METAL CORROSION

This is due to the chemical action of flowing liquid metal (eg. mercury) at high temperatures. The corrosion reaction involves either dissolution of a solid metal by a liquid metal. (or) liquid metal may penetrate into the solid metal.

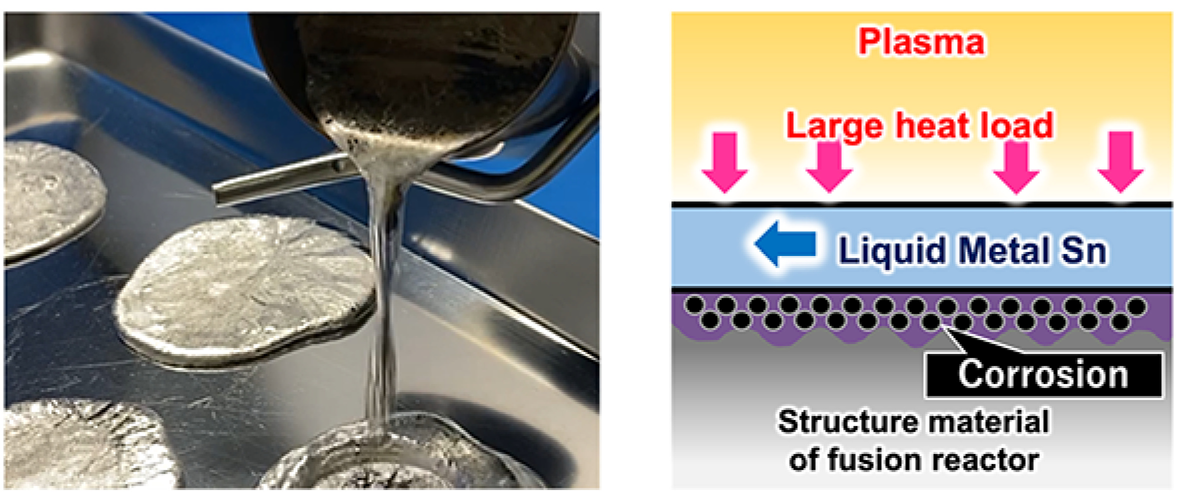

Figure 5. (Left) Liquid metal fluid, (Right) Mechanism of liquid metal divertor and corrosion issues.

DISCUSSION QUESTIONS

- Compare the stability of porous oxide layer, non porous oxide layer, volatile oxide layer with suitable reasons and examples
- Decarburization occurs at a high temperature 700 °C in steel and decreases the strength of the materials. Justify